PointNeuron: External Validation and Targeted Reconstruction Improvements

Revised research proposal for discussion with Dr. Arij

Meeting: 8 October 2026 | Revised: 7 October 2026

1 Motivation and current evidence

PointNeuron1.0 now provides a frozen reconstruction baseline and a sample-level diagnostic record. The next research phase should establish how this baseline transfers to a compatible external dataset, then test improvements against specific observed limitations. PointNeXt, a DETR-inspired proposal module and GATv2 remain candidate architectural directions; their benefits in this task have not yet been demonstrated.

The promoted pipeline converts a volume to foreground points, aggregates local guarded30 proposal predictions, selects graph nodes, constructs a foreground-geodesic graph and exports SWC. On the 63 evaluated stable-domain Gold166 samples (100-162), it achieved mean edge F1 0.8295, mean bridge count 1.444 and local point-distance SWC F1 0.8150. These are existing local results, not an external-test result or a direct comparison with the original paper benchmark. [1,2]

Diagnostic evidence	Research implication
Samples 7, 16, 23 and 24 still require 124, 104, 73 and 82 bridges with oracle nodes.	Proposal error alone cannot explain these cases. Foreground support and graph construction must be investigated together.
Sample 22: learned-node edge F1 0.110; oracle-node F1 0.890, with zero bridges.	Upstream node placement or selection matters; the same explanation should not be applied to every hard case.
Sample 19: segment coverage falls from 0.986 to 0.409 after selection.	Selection deserves a separate controlled comparison before replacing the proposal model.
Samples 3, 4 and 6: 17, 25 and 24 out-of-bounds annotated nodes.	Alignment/data-pairing problems must be separated from foreground preprocessing and reconstruction-model errors.

Research objective

Determine which combinations of input representation, proposal selection and edge prediction improve reconstruction geometry and annotation agreement without introducing unsupported fallback connections. The contribution is a tested explanation of failure mechanisms and validated improvements, rather than the integration of several newer architectures alone.

Status: PointNeuron1.0 diagnostic closure is complete. This document proposes future work for review; it does not report PointNeuron2.0 implementation, Allen experiments or newly measured performance.

2 Input representation and external validation

2.1 Preserve the actual patching protocol

Each dataset sample is a volume paired with a neuron annotation. Local spatial patches are formed within that volume, not across neurons. The current crop is cubic: radius 96 gives an interior width of 193 voxels at integer centers. Inference uses stride 96, a minimum of 256 foreground voxels and at most 256 coverage-selected patches. Actual counts in completed promoted runs were 3-220. Local proposals are aggregated in whole-volume coordinates before graph reconstruction. [1]

Training caches contain 512 points per patch. Current full-volume aggregation defaults to 2048 sampled input points per patch, but this count was not serialized in the saved inference archives. This configuration difference could be tested in a later controlled ablation if unexplained failures remain; no observed failure is attributed to it here.

2.2 Reframe skeleton-guided adaptive sampling as a hypothesis

The original proposal suggested using a classical skeleton to guide sampling. Retain this as an optional experiment, but call an automatically extracted skeleton a **reference skeleton or pseudo-label**, not ground truth. It may omit thin processes or contain erroneous connections. A sampling method trained to reproduce those errors could reinforce them.

Compare the existing sampling method with an image-derived, geometry-guided alternative while holding downstream components fixed. Evaluate terminal, branch and segment coverage before and after sampling, as well as runtime and memory. Any differentiable sampler should specify which parameters are learned, how gradients are obtained and how the sampling budget is enforced. Differentiability alone does not establish improved reconstruction.

Manually annotated SWC may supervise training and evaluate outputs, but must not guide sampling on held-out test neurons. An image-derived reference available at inference is a different input and must be documented. If manual correction is allowed, report it as a separate assisted protocol, with its intervention and time budget.

2.3 First milestone: Allen compatibility and frozen-model evaluation

Allen Cell Types is the proposed external dataset, conditional on recovering compatible paired microscopy stacks and reconstructions. Allen describes bright-field images of biocytin-filled cells, with dark cells on a light background, and provides access to image stacks, projections and SWCs. A projection or an SWC alone is not an equivalent input to the current volume-to-reconstruction pipeline. [4]

Before evaluation, verify stack completeness, coordinate conventions, voxel spacing, image/SWC alignment, intensity polarity and annotation scope. Define a reproducible input-conversion protocol without using held-out outcomes to tune it. Keep model weights and reconstruction settings frozen for the baseline test; report any later preprocessing or model adaptation separately.

Carry dataset identity, sample identity, species, brain region and cortical layer where recorded. Leave unavailable metadata null. Report exclusions and execution failures explicitly. This proposal does not presume an Allen-specific directory structure or promise transfer performance before compatibility checks.

3 Encoder and skeleton-proposal hypotheses

3.1 Candidate encoder: PointNeXt

PointNeXt remains a candidate replacement for DGCNN/EdgeConv, motivated by the possibility of improving multiscale geometric features. Its accuracy and efficiency on neuronal reconstruction must be measured rather than assumed. The original proposal's assertions of guaranteed speedup and superior reconstruction should be treated as hypotheses.

An encoder comparison should retain the same data splits, sampling protocol, proposal head and graph procedure wherever compatible. Document unavoidable interface changes. Compare proposal coverage, terminal coverage, downstream geometry/topology, parameter count, runtime and peak memory under a stated compute budget. Keep optional transformer fusion out of the initial comparison so its contribution can be tested separately.

3.2 Investigate selection before replacing proposal generation

The saved selection deficits show that good candidates can be lost before connectivity. First compare selectors using the same saved proposal pool and matched node budgets. Track candidate-to-selected coverage loss, skeleton precision, spacing and the final graph metrics. This experiment can distinguish limitations of the selection procedure from those of the proposal generator without retraining the model.

3.3 Candidate proposal head: DETR-inspired set prediction

A transformer decoder with learned queries could predict coordinates, objectness and radius as a set. This is a research alternative, not an established solution. Before implementation, define what a target node represents: annotated endpoints and branch points, centerline samples at a specified spacing, or an explicitly defined combination. Raw SWC sampling density should not silently determine the detection task.

Specify the maximum number of predictions per patch, treatment of unmatched queries, coordinate and radius units, and handling of targets that exceed the query budget. A fixed query budget must not recreate the coverage bottleneck already seen during node selection. Retain endpoint and branch coverage as explicit evaluation measures.

The original DETR formulation uses bipartite matching and a set-based objective to avoid conventional NMS in its detection setting. [3] For this application, test whether set prediction reduces reliance on suppression; do not claim that it automatically eliminates whole-neuron post-processing. Predictions from overlapping neuronal patches still require an explicit deduplication or reconciliation procedure.

3.4 Acceptance criteria for these alternatives

Promote an encoder or proposal change only when a matched comparison improves the intended coverage or geometry measure without an unacceptable deterioration in connectivity, bridge dependence or computational cost. Specify the trade-off and decision rule before inspecting held-out results. Better training loss alone is insufficient: the existing longer proposal run already demonstrated that loss and end-to-end reconstruction quality can disagree. [1]

The 512-versus-2048 point-count comparison, encoder replacement, proposal-head replacement and selector comparison should be separate ablations. Their outcomes determine whether a combined model is justified.

4 Connectivity and staged integration

4.1 Central hypothesis: learn edge reliability from several forms of evidence

Oracle-node failures motivate testing edge reliability and graph assembly under difficult foreground conditions. They establish that good nodes alone are insufficient in several cases, but do not isolate thresholding, image geometry and graph rules as independent causes. The existing connectivity GAE also underperformed the initializer; this does not establish that another learned graph module will succeed. [1]

Construct a documented candidate-edge set and compare a simple edge-scoring baseline with a GATv2-based alternative. Hold nodes, candidate edges and evaluation targets fixed when comparing scorers. Candidate features may include learned node features, Euclidean displacement, radius relations, endpoint confidence, directional continuation, snapping distance and foreground-geodesic support. Represent missing or unreachable support explicitly instead of interpreting a fallback cost as an observed path length.

Measure candidate-edge recall against the evaluation target before training an edge predictor: scoring cannot recover a target edge excluded from its candidate set. If multiple candidate scales are explored, separate their effect from the effect of the scoring network. GT-derived candidate sets or node sets are labeled oracle diagnostics, never the deployed inference input.

4.2 Separate edge prediction from final graph assembly

Compare edge quality before tree assembly with the exported reconstruction afterward. A bridge is a fallback connection added when constrained construction cannot naturally connect components; it is not a biological neuron branch. Some bridges have foreground support, while others use unreachable fallback connections. Evaluate their count jointly with reachability and annotation hit rate.

For future decoding experiments, report the trade-off between forcing a single connected tree and retaining disconnected but better-supported components. If SWC export requires a connected result, disclose the additional fallback edges and their evidence. A structurally valid tree can still contain wrong connections.

4.3 Annotation agreement is not independent biological validation

The current topology target is induced by mapping selected nodes to the annotated SWC and building connectivity from SWC path distances. Edge matching measures agreement with that induced morphology. Image support provides another diagnostic, but can also follow noise or adjacent structures. Neither proves independent biological plausibility.

If biological plausibility is claimed in future work, add a specified expert-review protocol or independent anatomical evidence. Report annotation agreement, foreground support, structural validity and independent review as distinct outcomes. Topology-aware penalties may encourage consistency; they do not guarantee biologically correct trees.

4.4 Training and integration strategy

Begin with frozen inputs and stage-specific comparisons: selection, candidate construction, edge scoring and decoding. Train an encoder/proposal alternative only when justified by the diagnostic evidence. Consider joint fine-tuning after individual benefits are established. State which operations are differentiable and which remain discrete; do not describe thresholding, hard selection and graph export as jointly differentiable merely because the neural modules support backpropagation.

5 Evaluation, milestones and meeting decisions

5.1 Evaluation protocol

Use whole-neuron training, validation and test splits; patches from the same neuron must remain in the same split. Where specimens or donors create related samples, group them appropriately. Use validation data for model choice and reserve held-out data for final comparisons. Describe existing Gold166 results as the evaluated baseline, without retrospectively relabeling them as an untouched external test.

Evaluation level	Required reporting
Data and execution	Alignment/data-pairing status; foreground thresholds and cap status; exclusions; completed and failed execution.
Proposals and selection	Candidate and selected node counts; segment, terminal and branch coverage; skeleton precision; coverage loss after selection.
Graph and reconstruction	GT-induced edge precision/recall/F1; reachability; bridge count and hit rate; SWC validity and roots.
Spatial geometry	Exported-SWC point-distance precision/recall/F1; explicitly labeled approximate ESA/DSA/PDS; distance thresholds, sampling step and coordinate units.
Reliability and resources	Per-sample results and uncertainty; stratified summaries where metadata permits; review attribution; runtime, peak memory and manual intervention.

Do not use an unspecified single "F1" for both topology and spatial fidelity. Keep the local approximate metrics distinct from official benchmark implementations. For external data, document unit conversion and spacing rather than assuming that a voxel threshold has the same physical meaning across datasets. Retain missing values and report denominators. The diagnostic labels CLEAR/REVIEW indicate review needs, not biological correctness or actual execution failure. [1,2]

5.2 Proposed milestones and decisions for Dr. Arij

Milestone A: Approve the Allen compatibility checklist and external-test protocol. Deliver a documented compatible cohort or a clear account of unavailable inputs.

Milestone B: Evaluate the frozen baseline and deliver the same per-sample diagnostics, including failures and metadata.

Milestone C: Choose one controlled intervention from the observed failure mechanisms; agree on comparators, primary metrics and promotion criteria before running it.

Milestone D: Combine only supported changes, then repeat held-out evaluation and any agreed independent morphology review.

The immediate meeting decisions are the external-validation scope, the first intervention to prioritize, and the evidence required to accept an improvement. PointNeuron2.0 is a proposed research direction pending that review; no performance gain is promised in advance.

References and evidence

[1] PointNeuron project. *PointNeuron1.0 Reimplementation and Evaluation*, diagnostic closure and schema cleanup, September 2026. Local report: docs/PointNeuron1_Report.pdf.

[2] Saved diagnostic outputs: tmp/diagnostics/pointneuron1_sample_diagnostics.json and pointneuron1_sources.json; schema: docs/sample_diagnostics.md. These preserve source-qualified observations and missingness.

[3] Carion et al. *End-to-End Object Detection with Transformers*, ECCV 2020. Original paper.

[4] Allen Institute. *Cell Types Database: API*, morphology and image access documentation. Official documentation. Consulted 7 October 2026.